Competent

58/60

97%

ELECTRICITY AND ELECTRONICS

SUMMATIVE ASSESMENTS

Module #	671101000-KM-03:
NQF Level	level 4,
Notional hours	130
Credit(s)	Cr 13
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

Welcome to your Summative Assessment!

This document deals with the summative assessment of your competence towards the outcomes which you need to achieve. "Summative Assessment is assessment for making a judgement about achievement. This is carried out when a learner is ready to be assessed at the end of a programme of learning" (SAQA: Criteria and Guidelines for Assessment Policy Document,).

Special arrangements for assessment

Place	Fastc Electrical
Language	English
Resources	
Barriers	N/A

People to be involved with assessment

Learner:	Sela Phiri	Manager:
Trainer:	Miss R. Nyakala	Mentor / Coach:
Assessor:		Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner:

Learners Name: Sela Phiri

Learner's signature: S. Phiri Date:

Assessors Name: Ruth Nyakala

Assessor's signature: Date:

ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____ Date: _____

Employee witness: _____ Date: _____

EASTC TECHNOCENTRIC VARSITY

5

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: Sela Phiri	Signature
Assessors Name:	Signature:

EASTC TECHNOCENTRIC VARSITY

LEARNING UNIT 1:

KM-03-KT01: Fundamentals of electricity (35%)

Learning Outcome 1:	Principles and fundamental concepts of electricity
Learning Outcome 2:	Definitions, types, properties and applications of conductors, insulators and semi-conductors
Learning Outcome 3:	Concepts, theories and principles of Electrical Circuits
Learning Outcome 4:	Calculations on basic electrical circuits (resistance, voltage and current) and power
Learning Outcome 5:	Basic principles and calculation of magnetism

Task	Questions Description	Marks
1	List types of materials used for conductors, insulators and semi-conductors and describe their mechanical and electrical properties and applications.	10

Conductors

- Copper
- Aluminium
- Silver, Gold and Brass

- Electrical properties
  - very low electrical resistance
  - Allow electric current to flow easily
  - High electrical conductivity
- Mechanical properties
  - Ductile (can be drawn into wires)
  - Malleable (can be shaped)
  - Strong and durable
  - Good heat conductor
- Applications
  - Wires Cables

Insulators

- Material used

5

Task	Questions Description	Marks
2	a) Name, describe and explain the different types of magnets and their properties b) Explain fundamental magnetic concepts by naming the five characteristics of magnetic lines of force and explaining the relationship between flux and flux density	10

Permanent magnets : These retain their magnetism for long periods without an external power source eg (Alnico)

- Temporary Magnets : These act like magnets only while within a magnetic Field.
- Electromagnets : These consist of a wire coil wrapped around a metal core. They are only magnetic when an electric current flows through the coil. Their strength can be easily controlled by changing the current

b. Fundamental magnetic concepts

* They alway form continuous, closed loops
* They never cross or intersect each other
* They travel from the north Pole to the South Pole outside the magnet
* They seek the path of least resistance (reluctance
* They act like stretched rubber bands, trying to shorten themselves

5

Task	Questions Description	Marks
3	Describe, with the aid of a drawing, the application of the right-hand grip rule and show how a magnetic field is established when an electrical current flows through a conductor, by using formula, calculate the force on a current carrying conductor.	10

Fleming's right hand Rule

~~Direction of Force~~ Current

Electric Current

$F = B \cdot I \cdot L \sin\theta$

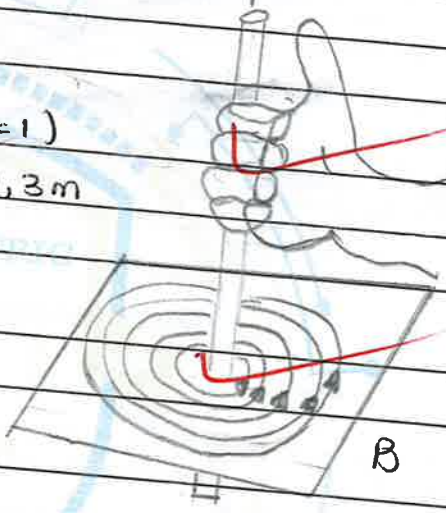
With $\theta = 90°$ ($\sin 90° = 1$)

~~Direction of Field~~

$F = 0{,}2\ T \times 5A \times 0{,}3m$

$= 0{,}3N$

~~Direction of Current~~



The right-hand Grip rule (also known as the Corkscrew Rule) is a simple mnemonic used to determine the direction of the magnetic field around a straight current-carrying wire

- The Thumb: Point your right thumb in the direction of the Conventional Current flow positive to negative
- The Fingers: Curl your fingers around the imaginary conductor
- The Field: The direction in which your fingers curl indicates the direction in which your fingers curl indicates the direction of the magnetic flux line (the magnetic field)

Establishing a magnetic field when DC current flows through a conductor, it doesn't just stay inside the wire it creates a magnetic field in the surrounding space. Concentric circles and field strenght and interaction

- To calculate this force,

$F = B \cdot I \cdot L \cdot \sin(\theta)$ F = Force acting on the Conductor

B = magnetic Flux I = Current L = Lenght

Feature	Earthing	Bonding
Connection	Connected to the Earth	Connected to each other
primary Goal	To trip the circuit breaker	To prevent a voltage difference
Focus	Electrical equipment casings	non-electrical metal/work (pipe/Frames

Regulatory requirement for earthing and Bonding

1. Continuity connection
   - Permanent Connections
   - Corrosion Resistance
   - Protection from damage
2. Resistance and impedance Limits
   - low Resistance Path: The earth-loop impedance must be low enough to allow a fault current of sufficient magnitude to flow
   - Automatic Disconnection
3. Resi sizing of Conductors
   - Cross-sectional Area
   - Main Bonding - Specific minimum diameters are required for main bonding Conductors
4. Testing and certification
   - Mandatory Inspection: No electrical installation can be legally passed or energized until the earthing and bonding have been tested

LEARNING UNIT 3:

KM-03-KT02: Electronics (30%)

Learning Outcome 1:	International Electro technical Commission (IEC) symbols for electronic components
Learning Outcome 2:	Principles, safety precautions, identification and basic function of electronic components
Learning Outcome 3:	Basic electronic circuits
Learning Outcome 4:	Principles of rectification, single phase and 3-phase

Task	Questions Description	Marks
6	a) Describe the various inductors used and typical applications b) Describe the construction and uses of various semiconductor devices c) Draw basic electronic circuits using International Electrotechnical Commission (IEC) symbols	10

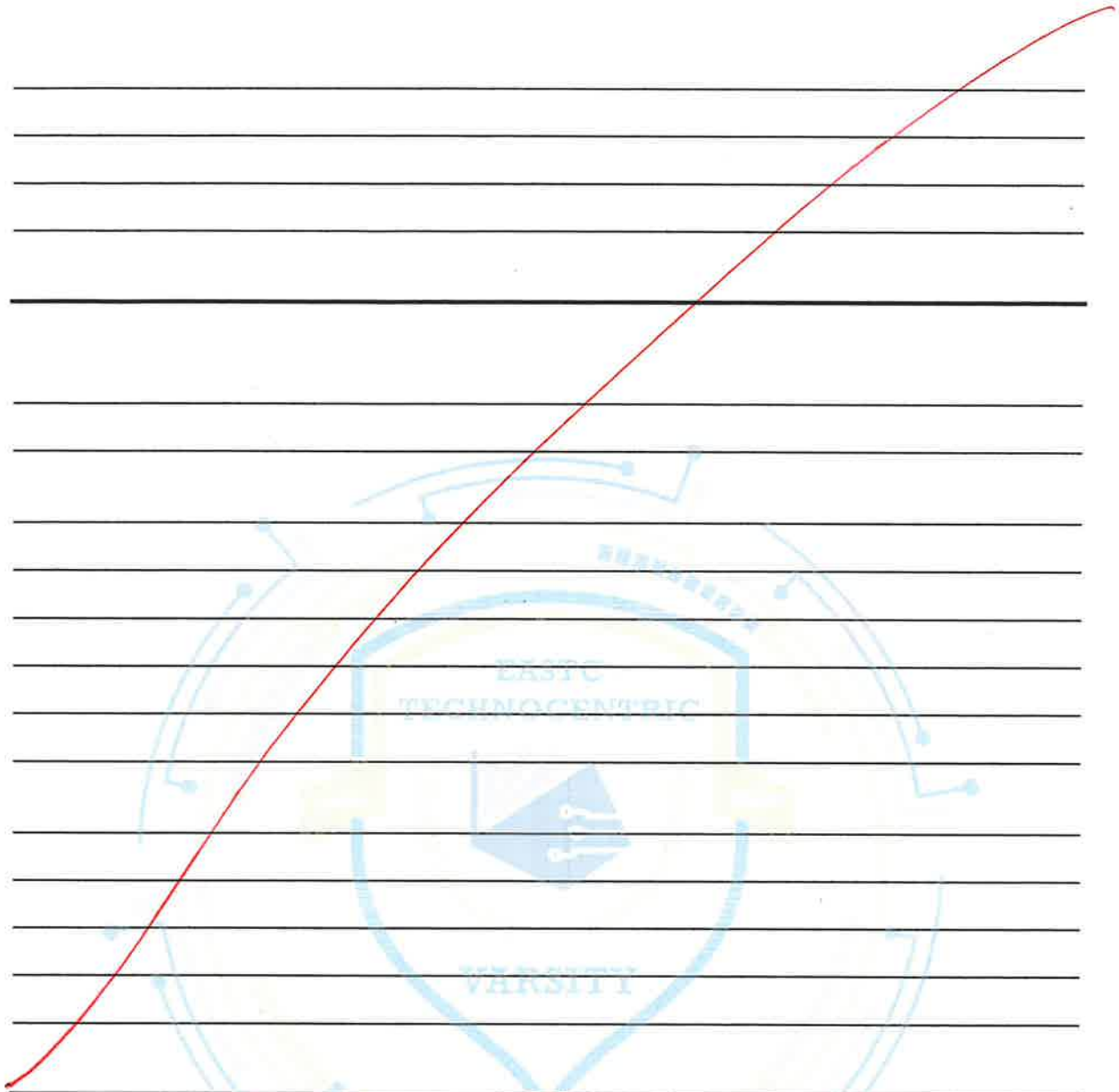
a. Inductors store energy in a magnetic field when electric current flows through them. Their performance depends largely on the core material used

Inductor type	Description	Typical Application
Air core	Coils wound on non-magnetic forms	high-frequency applications like RF tuning and radio transmitters

Construction and Uses of Semiconductor devices

* Semiconductor are the building blocks of modern electronics usually made from Silicon or Germanium doped with impurities
* Diode - Construction created by joining a P-type material (excess holes) and an N-type material to form a P-N junction

Uses: Rectification (AC to DC) Signal Clipping and reverse polarity protection



5

Assessor's Report

Candidate's Name	Sela Phiri	ID No.	9406111175089
Assessor's Name	RUTH NYAKALA	Reg. No.	
Title			

ASSESSMENT DECISION

Outcome	C	NYC	Comments

OVERALL COMPETENCE JUDGEMENT:

Comments	
Date	
Signature of Assessor	Signature of Candidate S. Phiri

5

Candidate Appeal Form

Candidate's Name:	ID No.	
Assessor's Name:	Reg. No.	
Title:		
Date:		
SECTION 1		
Candidate's reason for disagreeing with the assessment decision		
Assessor's rationale for the assessment decision		
Candidate's signature		
Assessor's signature		
SECTION 2		
Moderator's decision		
Moderator's signature		

EASTC TECHNOCENTRIC VARSITY

5

ASSESSMENT REVIEW

NAME of LEARNER Stla Phiri	NAME of ASSESSOR
VENUE East C	DATE of REVIEW

Review Dimension	ASSESSOR	LEARNER/ CANDIDATE	ACTION
The principles/criteria for good assessment were achieved?	☐Agree ☐Disagree	☐Agree ☐Disagree	
The assessment related to the registered unit standard?	☐Agree ☐Disagree	☐Agree ☐Disagree	
The assessment was practical?	☐Agree ☐Disagree	☐Agree ☐Disagree	
It was time efficient and cost-effective and did not interfere with my normal responsibilities?	☐Agree ☐Disagree	☐Agree ☐Disagree	
The assessment instruments were fair, clear and understandable	☐Agree ☐Disagree	☐Agree ☐Disagree	
The assessment judgements was made against set requirements	☐Agree ☐Disagree	☐Agree ☐Disagree	
The venue and equipment was functional?	☐Agree ☐Disagree	☐Agree ☐Disagree	
Special needs were identified and the assessment plan was adjusted	☐Agree ☐Disagree	☐Agree ☐Disagree	
Feedback was constructive against the evidence required	☐Agree ☐Disagree	☐Agree ☐Disagree	
An opportunity to appeal was given	☐Agree ☐Disagree	☐Agree ☐Disagree	
The evidence was recorded	☐Agree ☐Disagree	☐Agree ☐Disagree	

LEARNER'S DECLARATION OF UNDERSTANDING

I am aware of the moderation process and understand that the moderator could declare the assessment decision invalid

S.		
Learner **Date**	**Assessor** **Date**	**Moderator** **Date**

5